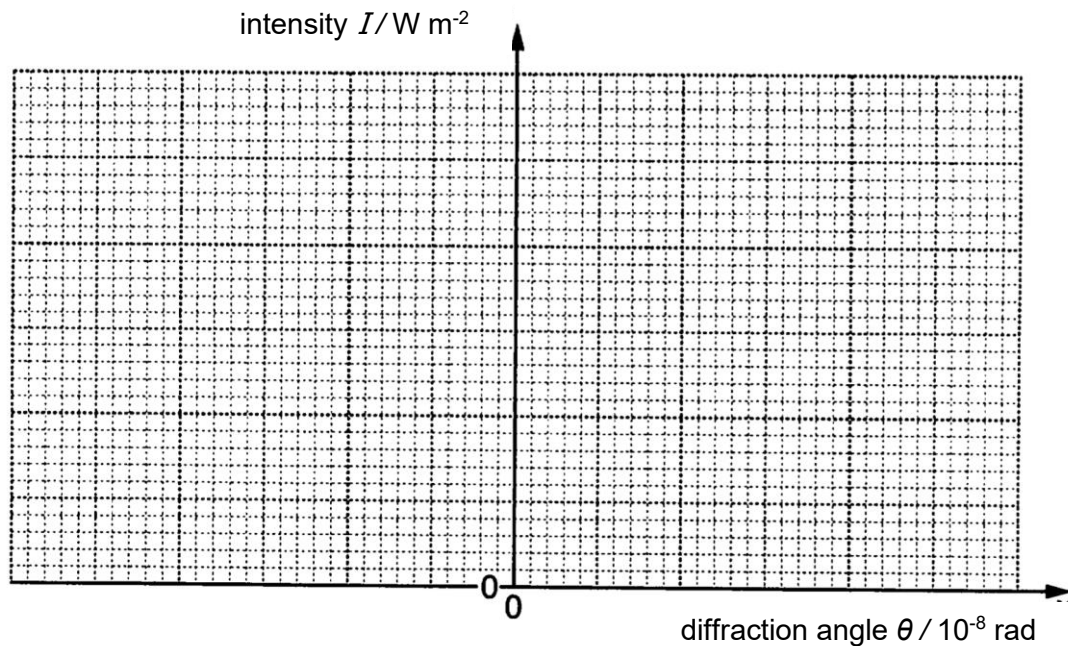


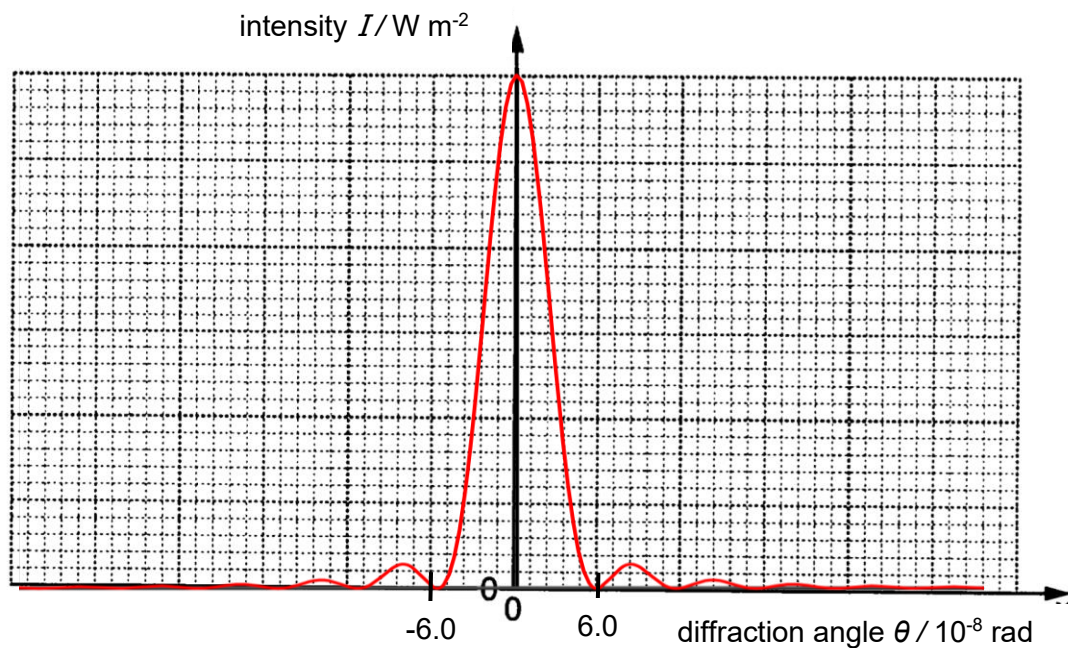
4	(a)	A Keck telescope at Mauna Kea, Hawaii, is the world's largest optical telescope and has a diameter of 10 m and is set to detect waves of wavelength 600 nm. The distance, in metres, from the aperture to the viewing screen is L .	
	(i)	Calculate the diffraction angle θ_{min} at which the first minimum of the diffraction pattern is observed on the viewing screen.	
		$\theta_{min} = \dots\dots\dots$ rad	[2]
		Solution:	
		$\sin \theta_{min} = m \frac{\lambda}{b}$	1
		$\sin \theta_{min} = (1) \frac{600 \times 10^{-9}}{10}$	
		$\theta_{min} = 6.0 \times 10^{-8} \text{ rad}$	1
	(ii)	Determine, in terms of L , the width of the central bright fringe of the diffraction pattern observed on the viewing screen.	
		width = $\dots\dots\dots$ m	[2]
		Solution:	
		$\tan \theta_{min} = \frac{y}{L}$	
		$\tan 6.0 \times 10^{-8} = \frac{y}{L}$	1
		$y = (6.0 \times 10^{-8})L$	
		$\text{Width} = 2y = 2(6.0 \times 10^{-8})L = 1.2 L \times 10^{-7} \text{ m}$	1

- (iii) On Fig. 4.1, sketch a graph to show the variation with diffraction angle θ from the central maximum of the intensity I of the light on the viewing screen. Include the angles for the first minima.



[3]

Fig. 4.1



1 mark:

Smooth diffraction pattern curve, with 2 maxima on either side. 5 maxima in total when central maxima is included. Show to 3rd minima [3 maxima are insufficient to show a 'trend'!]

1 mark:

Exact values of intensity axis labelling with values are not required, however the value for the 1st minima should be included.

1 mark:

		<i>(Other details)</i> -Width of central fringe must be twice that of the other maxima fringes. -Minima must touch zero intensity. -Maxima intensity must reduce on either side as it gets further from central maxima fringe. (Intensity of central maxima must be shown to be significantly higher than the higher order maxima.)	
	(b)	A radio telescope at Arecibo, Puerto Rico, has a diameter of 305 m and is designed to detect radio waves of wavelength 0.75 m.	
	(i)	State two physical quantities that determine the resolving power of the telescope.	
		quantity 1 : quantity 2 :	[2]
		Solution: quantity 1: size/diameter of the aperture quantity 2: wavelength of the electromagnetic wave	1 1
	(ii)	Explain quantitatively whether the Keck telescope used in the detection of light waves has a higher or lower resolving power compared to the radio telescope used in the detection of radio waves at Arecibo.	
			
			[2]
		Solution: Keck Telescope: $\theta_{\min} \approx \frac{\lambda}{b}$ $= \frac{600 \times 10^{-9}}{10}$ $= 6.0 \times 10^{-8} \text{ rad}$ <i>(No marks awarded if candidates substitute wavelength as 0.75 m)</i> The Keck telescope as stated in (a) is set to detect waves of wavelength 600 nm Radio Telescope: $\theta_{\min} \approx \frac{\lambda}{b}$ $= \frac{0.75}{305}$ $= 3.0 \times 10^{-3} \text{ rad}$	

			Since the minimum angle for the radio telescope is much larger than that of the Keck telescope, it indicates that the Keck telescope is able to observe objects as separate images even though the objects are much closer to each other, whereas the radio telescope would not be able to pick up the distinction of the objects due to its larger limiting angle. Thus, the Keck telescope has a higher resolving power.	1 1
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5	(a)	Define potential difference between two points in a circuit	
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